CMPE180A — Dr Gomez — Spring 2026
Sample Exam 1 — Explicit Version

Practice instructions: These sample questions are written in a more explicit style. Read each prompt carefully, pay attention to the requested function behavior, and use the sample input/output to guide your interpretation.

1. Q1. Reverse a Linked List (Iterative and Recursive)

Write two C++ functions for a singly linked list:

1) Node* reverseIterative(Node* head)

2) Node* reverseRecursive(Node* head)

Both functions must return the head of the reversed list.

Sample input: 4 -> 7 -> 1 -> 9

Expected output: 9 -> 1 -> 7 -> 4

State the time complexity and extra space complexity of each version.

2. Q2. Recursive Fibonacci with Memoization

Write a recursive function int fibMemo(int n, vector<int>& memo) that computes Fibonacci numbers using memoization.

Use the convention fib(0)=0 and fib(1)=1.

Sample input: n = 7

Expected output: 13

Also explain why this version is more efficient than the naive recursive Fibonacci implementation.

3. Q3. Balanced Parentheses Using a Stack

Write a function bool isBalanced(const string& s) that returns true if all (), {}, and [] symbols are balanced and properly nested.

You must use a stack.

Sample input 1: "{[()()]}"

Expected output 1: true

Sample input 2: "{[(])}"

Expected output 2: false

Briefly explain what is pushed onto the stack and when items are popped.

4. Q4. Backtracking: Generate All Permutations of a String

Write a backtracking algorithm that prints all permutations of a given string.

You may assume all characters are distinct.

Sample input: "ABC"

Expected output: ABC, ACB, BAC, BCA, CAB, CBA

Show the first two levels of the recursion tree or describe the first few recursive calls clearly.